

Mehmet Turan Yardımcı

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Research Interests

Online reinforcement learning fine-tuning of vision-language-action (VLA) policies; continually self-improving robots that keep their skills while they keep learning; humanoid loco-manipulation.

Education

Karlsruhe Institute of Technology (KIT)

M.Sc. Computer Science

Karlsruhe, Germany

Oct 2026 –

Çukurova University

B.Sc. Computer Engineering (English-medium)

Adana, Türkiye

2021 – 2025

- Thesis: *Benchmarking Local Path Planners in ROS using the BARN Dataset*, supervised by Dr. Yunus Emre Coğurcu.
- Honor Student (Fall 2023–24), High Honor Student (Fall 2024–25).

Białystok University of Technology

Erasmus+ exchange semester, Computer Science

Białystok, Poland

Sep 2024 – Feb 2025

Publications and Preprints

- **M. T. Yardımcı**, Y. E. Coğurcu. *Uncertainty-Gated Exploration Noise Suppresses Task Collapse in Online RL Fine-Tuning of a Flow-Matching Vision-Language-Action Policy*. arXiv:2609.28838, 2026. Under review; also submitted to the CoRL 2026 Workshop on Continually Self-Improving Robots (non-archival).
- **M. T. Yardımcı**. *Critic Architecture Matters: Dual vs. Unified Critics for Humanoid Loco-Manipulation*. ICRA 2026 Workshop on Reinforcement Learning in the Era of Imitation Learning (RL4IL), Vienna, 2026 (poster). arXiv:2606.11891.
- **M. T. Yardımcı**, Y. E. Coğurcu. *Benchmarking Local Path Planners in ROS using the BARN Dataset*. Under review at the Journal of the Faculty of Engineering, Çukurova University. Code on GitHub.

Research Projects

Online RL fine-tuning of a flow-matching VLA policy

2026

Independent research

- Built a PPO fine-tuning pipeline for SmolVLA (450M parameters) on LIBERO-10 in the stochastic (Flow-SDE) sampling regime, ported from the RLinf reference recipe.
- Designed an uncertainty-gated exploration-noise controller that reallocates noise across task streams from random-network-distillation novelty and critic-ensemble competence signals, without task labels.
- Ran a matched-budget three-arm, three-seed campaign; the controller showed no pooled task collapse in any seed, against two of three seeds with fixed noise and three of three with learned noise.
- Built measurement tools for per-task collapse under four definitions, rescoring noise and instrument tares; measured that 96.02% of the action expert's elements stay bit-identical under the reference in-place bfloat16 update.

Dual vs. unified critics for humanoid loco-manipulation

2026

Unitree G1, NVIDIA Isaac Lab

- Built a multi-stage PPO curriculum for G1 whole-body locomotion (99.6% locomotion success), then trained dual- and unified-critic loco-manipulation policies on top and a standardized reaching benchmark; checkpoints are public on Hugging Face.

Hierarchical VLM planner with RL skills for the Unitree G1

2025 – 2026

NVIDIA Isaac Lab | github.com/mturan33/isaac-g1-hierarchical

- A vision-language planner (Qwen3-VL) sequences RL-trained skills (walk, reach, grasp, pull drawer, pick-and-place) from natural-language instructions; 6/6 drawer and 8/8 pick-and-place trials succeeded in simulation.

RL-to-imitation-to-VLA pipeline for the Unitree G1

2026 – present

NVIDIA Isaac Lab, LeRobot

- Collects demonstrations from trained PPO experts into LeRobot datasets and distils them into visuomotor policies (ACT, Diffusion Policy) toward GR00T N1.6 fine-tuning.

Reinforcement learning from scratch

2025

- Implemented PPO and SAC in NumPy and PyTorch for MuJoCo Ant, and PPO for ANYmal-C quadruped locomotion in Isaac Lab (17,000+ environment steps per second on a single RTX 5070 Ti).

Service

Reviewer, CoRL 2026 Workshop on Continually Self-Improving Robots

2026

Leadership

Team Captain, Alkar UAV Team, Çukurova University

Adana, Türkiye

TEKNOFEST Combat UAV category

2022 – 2024

- Led a team of 10+ members building autonomous UAV systems; integrated YOLO-based detection on Jetson with Pixhawk flight control.

Experience

Software Engineering Intern, Medcem and Kıvanç Tekstil

Adana, Türkiye

In-house software development in .NET and C#

Jul – Sep 2025

Skills

Robot learning: PPO, SAC, flow-matching policies, reward shaping, domain randomization

Simulation and frameworks: NVIDIA Isaac Lab, MuJoCo, LIBERO, LeRobot, RSL-RL, ROS, Gazebo

Programming: Python, PyTorch, C/C++, CUDA, Linux, Git

Languages: Turkish (native), English (C1; TOEFL iBT MyBest 110/120, C1 in all four sections)